

Measuring Tree Canopy Height using SAR Interferometry

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Outline

- 1 Introduction
- 2 Background: SAR Interferometry
- 3 Data Procurement
- 4 SNAP: Preprocessing Steps
- 5 Results, Diagnostics and Recommendations
- 6 References

Motivation

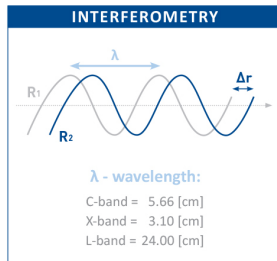
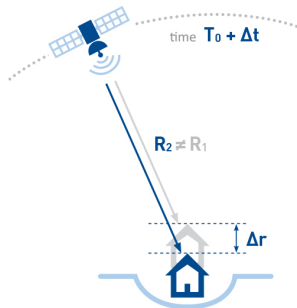
- Forest canopy height is important for biomass estimation, carbon accounting and habitat assessment.
- Remote sensing lets us capture wide-area images again and again. Using InSAR, we can use the phase information of radar signals to calculate height or elevation.
- Our focus: derive canopy height (average per-patch) using **Sentinel-1 SLC** via **SAR interferometry** (InSAR).
- Tools: ESA SNAP for SAR processing, QGIS for GIS operations.

Objectives

- 1 Present end-to-end reproducible workflow from Sentinel-1 SLC to canopy height maps (CHM).
- 2 Explain SNAP processing steps and mathematical justification used in height inversion.
- 3 Demonstrate QGIS post-processing (DSM-DEMs subtraction) and discuss observed error sources.
- 4 Discuss the obtained results along with final conclusions and key recommendations for improving accuracy.

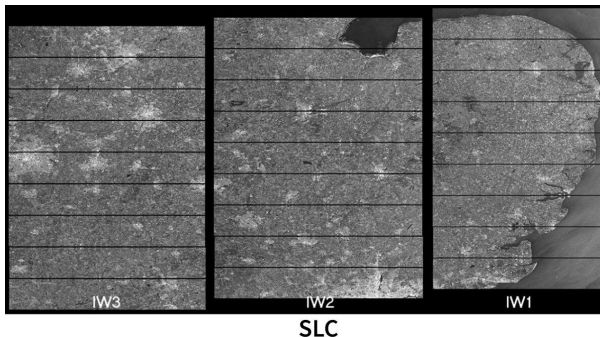
SAR Interferometry

- SAR (Synthetic Aperture Radar) interferometry is a remote sensing technique that combines two or more SAR images of the same area to create maps of surface topography or deformation.
- InSAR (Interferometric Synthetic Aperture Radar) uses the phase information from two or more SLC (Single Look Complex) SAR images to detect ground deformation or map topography with high precision.



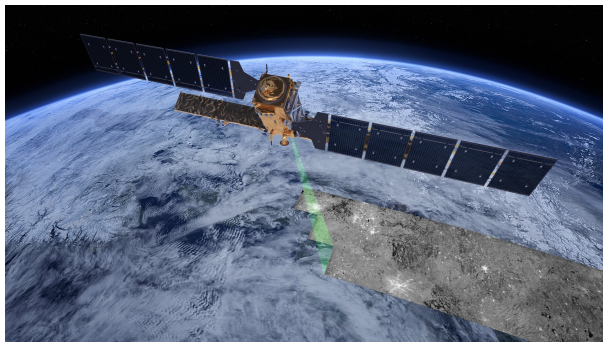
Why SLC (Single Look Complex)?

- SLC preserves preserving phase and amplitude information required for interferometry.
- SLC processing requires more storage/computation but preserves geometry and phase.
- SLC: complex values $s(\mathbf{r}) = A(\mathbf{r})e^{j\phi(\mathbf{r})}$.
 A = amplitude (strength of backscattered signal).
 ϕ = phase (angle of the backscattered signal).



Sentinel-1

- **12-day repeat cycle:** Provides frequent revisits, enabling coherent image pairs for deformation and topography.
- **Free global data:** Open-access SAR data ensures continuous, consistent time-series analysis.
- **C-band SAR:** Suitable wavelength for capturing ground deformation, vegetation structure, and surface changes.



Interferometry: Basic Idea

- Interferogram formed by multiplying master SLC with complex conjugate of slave:

$$I(\mathbf{r}) = s_1(\mathbf{r}) s_2^*(\mathbf{r}) = A_1 A_2 e^{j(\phi_1 - \phi_2)}.$$

- Phase difference encodes range difference between two acquisitions:
 $\Delta\phi(\mathbf{r}) = \phi_1 - \phi_2.$

Phase to Height: Mathematical Relation

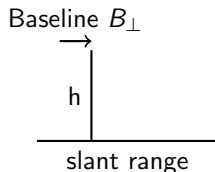
The interferometric phase difference due to terrain height is:

$$\Delta\phi = \frac{4\pi}{\lambda} \frac{B_{\perp} h}{r \sin \theta}$$

This shows that phase change is directly proportional to height difference.

where:

- λ = radar wavelength
- $B_{\perp}$ = perpendicular baseline
- h = height (elevation difference)
- r = slant range
- θ = incidence angle



Height Inversion

From the relation:

$$h = \frac{\lambda r \sin \theta}{4\pi B_{\perp}} \Delta\phi$$

- Estimate the unwrapped phase difference ($\Delta\phi$).
- Compute local incidence angle, slant range, and perpendicular baseline from orbital metadata.
- Convert the residual phase into absolute elevation using the above relation.

Coherence and Phase Quality

- Coherence γ measures the phase stability between two SLC acquisitions:

$$\gamma = \frac{|\langle s_1 s_2^* \rangle|}{\sqrt{\langle |s_1|^2 \rangle \langle |s_2|^2 \rangle}}$$

- High coherence (close to 1) $\rightarrow$ good, stable interferometric phase.
- Low coherence (close to 0) $\rightarrow$ decorrelated, unreliable phase.

Sources of Phase Error

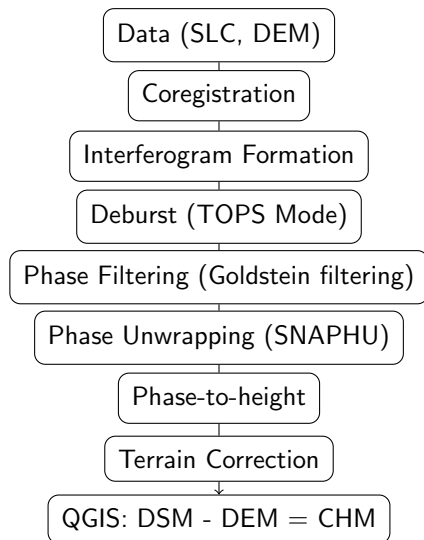
- Atmospheric delays.
- Baseline and geometry estimation errors
- Phase unwrapping errors
- Temporal decorrelation (vegetation growth, wind)

Study Area: IIT Dharwad

- Location: IIT Dharwad campus forest region
- Densely forested



Workflow Flowchart



- Sentinel-1 Level-1 SLC (master/slave pair) - Copernicus Open Access Hub.
- Download SLC pairs
- Master image:
S1A_IW_SLC__1SDV_20251029T005607_20251029T005635_061633_07B319_B413.SAFE
- Slave image:
S1A_IW_SLC__1SDV_20251017T005607_20251017T005635_061458_07AC11_93AF.SAFE
- Important: master and slave images must be from same sub-swath.

DEM Source

- Digital Elevation Model (DEM) obtained from **OpenTopography** portal.
- Freely available, void-filled, and easy to download through OpenTopography.



1. Coordinates

Horizontal Coordinates: WGS84 [EPSG: 4326]

Vertical Coordinates: WGS84 Ellipsoid/EGM2008 Geoid [EPSG: 3855]

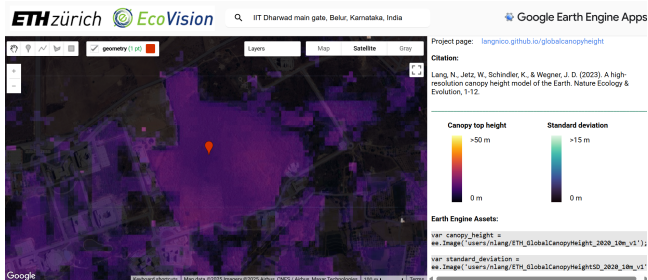
Units: meter

Data Selection Coordinates (Select an area on map)

$X_{min} = 74.93568360138944$ $Y_{min} = 15.483723788760884$ $X_{max} = 74.94566813654008$ $Y_{max} = 15.491706462404125$

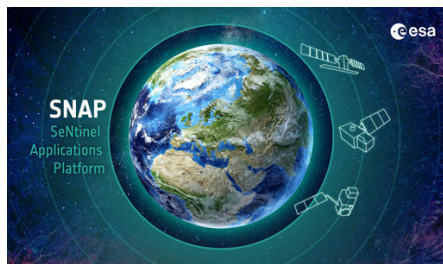
Global Canopy Height Reference (2020)

- Canopy height information obtained from the **Global Canopy Height 2020** dataset (Google Earth Engine Application).
- Source link:
<https://nlang.users.earthengine.app/view/global-canopy-height-2020>
- Dataset provides global tree canopy height derived using **GED1 + Landsat + Sentinel-2** fusion models.



SNAP: Overview

- **SNAP (Sentinel Application Platform)** is ESA's official, open-source software suite for processing Sentinel-1, Sentinel-2, and Sentinel-3 mission data.
- Designed specifically for **SAR InSAR**, and **ocean** data workflows, supporting SLC, GRD, and all Sentinel-1 IW, EW, SM modes.
- SNAP toolchain (Sentinel Application Platform) provides dedicated modules: Apply Orbit, TOPS processing, Co-registration, Interferometry, Unwrapping.

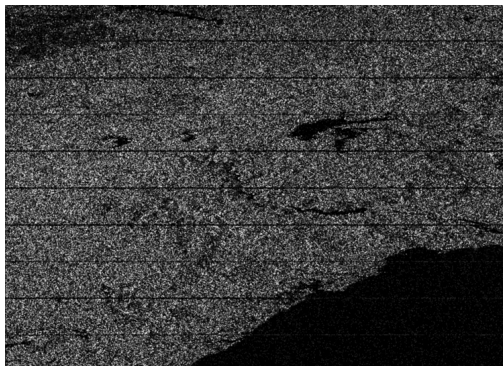


- **QGIS (Quantum GIS)** is an open-source, cross-platform Geographic Information System widely used for analysis, visualization, and raster processing.
- Provides a powerful graphical interface for working with raster data (DSM, DEM, CHM)
- Fully compatible with InSAR output from SNAP can open DSM/DEM layers



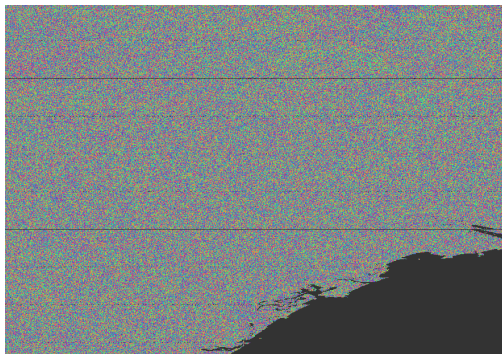
Step 1: Coregistration

- Aligns the master and slave SLCs pixel-by-pixel.
- Sentinel-1 TOPS mode needs the two radar images to be matched very precisely in both directions (range and azimuth) for correct processing.
- Ensures both acquisitions observe the same ground resolution cell.
- Even a 0.5-pixel misregistration can destroy interferometric phase.



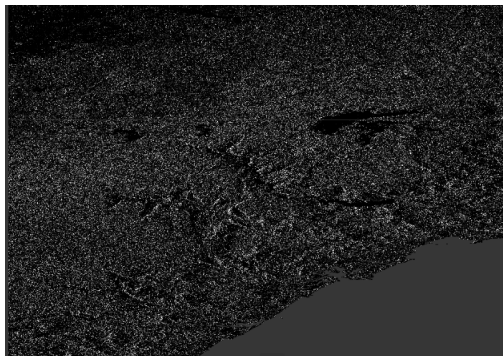
Step 2: Interferogram Formation

- Generates the complex interferogram by multiplying Master with the conjugate of Slave.
- Encodes phase information related to height, deformation, atmosphere, and orbit errors.
- The interferogram is wrapped ($-\pi$ to $+\pi$) and still contains flat-earth and topographic phase.
- Coherence is computed simultaneously to assess phase quality.



Step 3: Deburst (TOPS Mode)

- Merge bursts of TOPS IW mode into a continuous image.
- Removes azimuth discontinuities.
- Stitching based on precise timing and Doppler.
- Needed before filtering, unwrapping, or terrain correction.

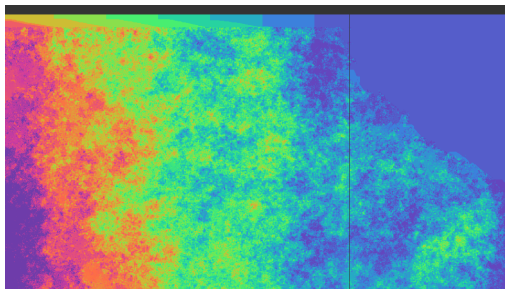


Step 4: Phase Filtering (Goldstein filtering)

- Goldstein filtering reduces high-frequency phase noise while preserving fringes.
- Improves fringe clarity and stabilizes the unwrapping process.
- Filtering enhances coherence $\rightarrow$ smoother phase for DSM generation.

Step 5: Phase Unwrapping

- Convert wrapped phase ($-\pi$ to π) to continuous phase.
- It is most crucial and time taking process
- SNAPHU uses network-flow optimization.
- Unwrapped phase is required to convert phase $\rightarrow$ height.
- Critical step — errors here propagate directly into height bias



Step 6: Phase to Elevation

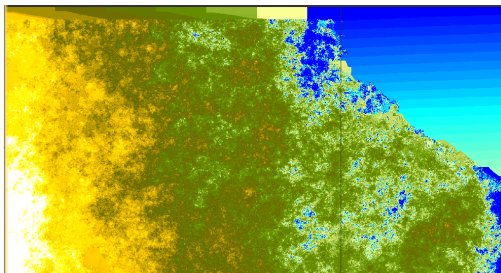
- After phase unwrapping, the phase still contains terrain-related information.
- SNAP uses the DEM to estimate the topographic phase and remove it, leaving only the residual height phase.
- This residual phase is converted into elevation using the radar geometry:

$$h = \frac{\lambda r \sin \theta}{4\pi B_{\perp}} \phi_{\text{residual}}$$

- Here, height depends on wavelength, baseline, slant range, and incidence angle.
- Output is a **radar-coded Digital Surface Model (DSM)**

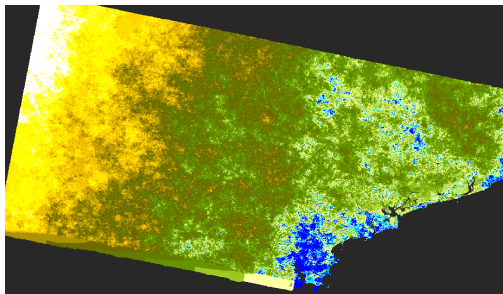
Step 7: Terrain Correction

- Converts the radar-coded DSM (slant-range geometry) into map coordinates (lat-long).
- Corrects geometric distortions such as foreshortening and layover.
- Applies geoid correction to convert ellipsoidal heights to orthometric heights.



Step 8: Export Final DSM

- Export the terrain-corrected elevation band as a GeoTIFF.
- The exported product contains:
 - Digital Surface Model (DSM)
 - Units in meters
 - Correct map projection
- This DSM is ready for further analysis, CHM generation.



QGIS: Preprocessing Steps for DSM and DEM

Goal: Prepare DSM and DEM so that the CHM (DSM – DEM) subtraction is accurate.

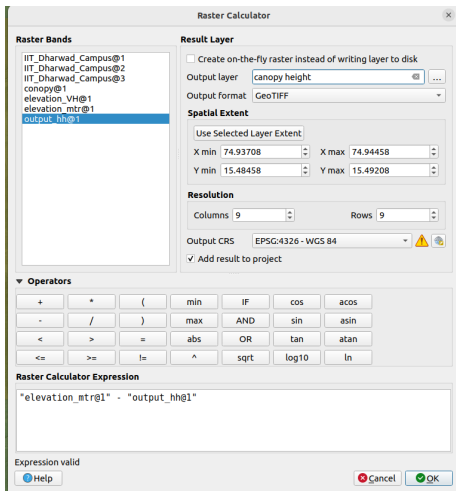
Steps:

- **Check CRS:** Ensure both DSM and DEM use the same projection.
- **Align Rasters:**
 - Match pixel size and grid alignment
 - Match raster extent
 - Guarantee cell-to-cell subtraction
- **Clip to AOI:** Limit rasters to the IIT Dharwad forest boundary for faster processing.

Computing Canopy Height Model (CHM) in QGIS

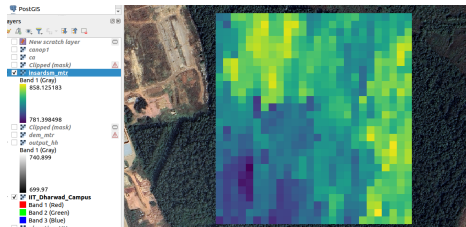
Mathematical Basis:

$$\text{CHM} = \text{DSM}_{\text{InSAR}} - \text{DEM}_{\text{Ground}}$$

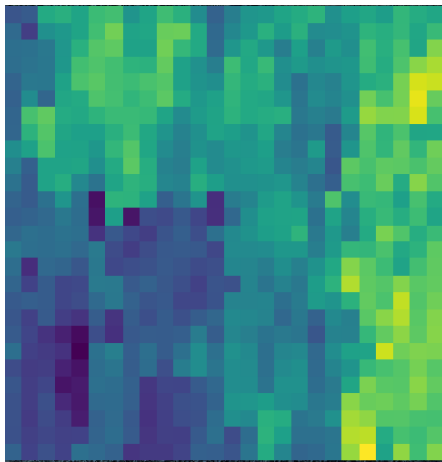


CHM Post-processing

- Apply heatmap color ramp and classify canopy height ranges.



InSAR-derived DSM (Elevation including canopy)

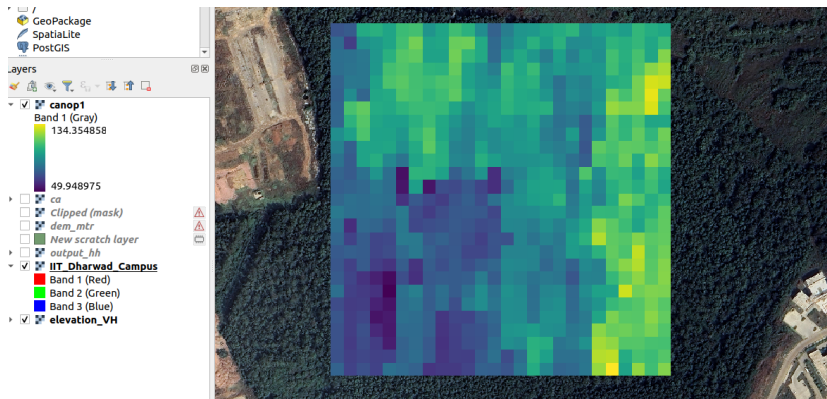


$$\text{Canopy Height Model (CHM)} = \text{DSM} - \text{DEM}$$

Observed statistics and Results

- Compute mean for CHM over Area of Interest.

Region	Expected (m)	Obtained (m)	Notes
IIT Dharwad AOI	16–20	60–80	Overestimation observed



Why did we get 60–80 m? (Likely Causes)

- 1 **DEM mismatch:** If SRTM DEM contains canopy bias or the DEM is misaligned.
- 2 **Phase unwrapping errors:** Low coherence leads to wrong unwrapping producing large height jumps.
- 3 **Incorrect baseline values:** Using wrong perpendicular baseline in conversion gives scaling error.
- 4 **Geocoding errors:** Pixel registration issues produce systematic offsets.

Possible Missing Steps in the Paper (Hypothesis)

- Paper may have used ICESat-2 corrections and particular GCPs to correct DEM bias (not fully documented).
- They might have used multi-pair stacking/temporal averaging to reduce noise.
- Use of polarimetric corrections or additional filtering (not fully described).

- **Use Multi-Pair InSAR Stacking:** Process several interferometric pairs and average the resulting DSMs to reduce phase noise, atmospheric effects, and random errors.
- **Adopt Longer-Wavelength or Active Sensors:** Explore L-band SAR (e.g., ALOS-2) or airborne/spaceborne LiDAR for improved vegetation penetration and validation.
- **Enhance Landcover Masking:** Use Sentinel-2 NDVI or ESA WorldCover to exclude non-forest areas before CHM extraction for cleaner canopy height outputs.

Alternative Sensors and Fusion

- L-band SAR (ALOS-2): better penetration for structural heights.
- Spaceborne LiDAR (ICESat-2, GEDI): direct canopy height measurements for calibration.
- Multisensor fusion (Sentinel-1 + Sentinel-2 + LiDAR) improves results.

Difficulties and Challenges Faced

- **Low Coherence Over Forest Areas** Dense vegetation causes decorrelation, making interferogram quality poor and generating unreliable phase values.
- **Phase Unwrapping Errors** Low-coherence regions produce residues and branch cuts, leading to incorrect unwrapped phase and meter-scale height biases.
- **High Computational Load** TOPS co-registration, interferogram formation, filtering, and SNAPHU unwrapping require significant RAM and disk space; large intermediate files slow down the workflow.
- **Time-Consuming Processing** Debursting, ESD, unwrapping, and terrain correction steps are computationally intensive and take long processing times.

Conclusions

- We used only **Sentinel-1 Level-1 SLC** data and followed the workflow described in the reference research paper to derive canopy height from InSAR.
- Although the full processing chain (SLC $\rightarrow$ interferogram $\rightarrow$ DSM $\rightarrow$ CHM) was implemented successfully, the final canopy heights were **not accurate** for the IIT Dharwad forest region.
- Additional corrections such as ICESat-2 ground truth, multi-pair stacking, or longer-wavelength SAR (L-band) are required to achieve reliable canopy height results.

References

-  Yi Wu et al., 2025. Estimation of Subtropical Forest Aboveground Biomass Using Active and Passive Sentinel Data with Canopy Height. *Remote Sensing*, 17, 2509.
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-  Practical InSAR Processing Tutorials (ESA / UNAVCO / SNAP Community Resources).
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-  Copernicus Open Access Hub – Sentinel-1 SLC Data Download. <https://dataspace.copernicus.eu>
-  OpenTopography – SRTM / Global DEM Access Portal. <https://opentopography.org>
-  Lang, N., Jetz, W., Schindler, K., & Wegner, J. D. (2023). Global Canopy Height 2020 (GEE Application). <https://nlang.users.earthengine.app/view/global-canopy-height-2020>

Thank You!